Chenfeng Li

Fargo, ND, 58102 | (872)-215-0270 | cfli@chenfengli.com | https://ChenfengLi.com

PROFESSIONAL SUMMARY

Data Scientist and Data Engineer with **3+ years of experience** designing and delivering scalable, end-to-end data solutions across machine learning, data engineering, and business intelligence. Highly proficient in **Python, SQL, R**, and advanced analytics, with extensive experience building **robust ETL pipelines**, distributed data processing systems, and cloud-native data architectures using **AWS, Azure, Google Cloud Platform, Snowflake, Apache Spark, and Databricks**.

Demonstrated expertise in leveraging modern **AI/ML technologies**, including deep learning, natural language processing (**NLP**), and Generative AI. Experienced with frameworks and tools such as **Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers,** and **OpenAI APIs** to develop predictive models, recommendation systems, and Language-based applications. Strong hands-on experience in feature engineering, model optimization, and deploying machine learning pipelines into production environments.

Skilled in data modeling, data warehousing, and implementing data governance, security, and quality frameworks. Proficient in workflow orchestration and real-time data processing using **Apache Airflow and Apache Kafka**. Adept at building interactive dashboards and reports using **Power BI, Tableau, and DAX**, enabling data-driven decision-making. Strong foundation in statistical analysis, time series forecasting, and algorithmic problem-solving, with the ability to translate complex data into actionable insights.

TECHNICAL SKILLS

- **Programming Languages**: Python, SQL, R, DAX, C, C++
- **Databases & DWarehousing:** MySQL, PostgreSQL, Microsoft SQL Server, AWS RDS, Snowflake, Azure Synapse Analytics, Google BigQuery
- **Cloud Platforms**: AWS (S3, Glue, RDS, DMS, CloudWatch, CloudFormation, IAM, SNS), Azure (Data Factory, Databricks, Synapse Analytics), Google Cloud Platform (GCP) (BigQuery, Cloud Storage)
- **Data Engineering & Processing**: Apache Spark, PySpark, Apache Airflow, Apache Kafka, AWS Glue, Azure Data Factory
- **Data Visualization & Reporting:** Power BI, Tableau, Excel, Power Query, DAX, Python (Matplotlib, Seaborn)
- **Libraries & Frameworks:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, PySpark, Boto3, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, NLP Toolkits, Generative AI APIs
- **Machine Learning & AI:** Regression (Linear, Ridge, Logistic), Decision Trees, Random Forest, Ensemble Models (XGBoost, LightGBM), Neural Networks (DNN, CNN), Object Detection (YOLO), Semantic Segmentation (DeepLabv3), Natural Language Processing (NLP) (text preprocessing, sentiment analysis), Feature Engineering, Model Optimization, Generative AI
- **Statistical Analysis:** Generalized Linear Models (GLM), Bayesian Inference, Time Series Analysis, Non-Parametric Methods

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate
- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Data Engineer Associate
- Microsoft Office Specialist: Excel 2019 Associate
- Google Advanced Data Analytics (Coursera)
- Deep Learning Specialization (DeepLearning.AI, Coursera)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern
***Packet Digital**, Fargo, ND | October 2023 – Present*

Weed Detection System Optimization for Drone-Based Agricultural Analytics (Polli) ***January 2026 – Present***

Redesigned the **end-to-end data and modeling pipeline** for high-resolution aerial weed detection in **Polli,** introducing scalable human-in-the-loop annotation and optimizing small-object detection, resulting in improved model generalization and detection performance.

Key Contributions:

- Diagnosed dataset bias (single-location sampling) and proposed a **multi-field data acquisition** strategy to improve spatial generalization and reduce overfitting.
- Designed a human-in-the-loop annotation system using **Label Studio** and **AWS,** enabling biologists to refine model-generated bounding boxes **without direct access to raw data.**
- Built an event-driven data pipeline using AWS Lambda to convert proprietary annotations into Label Studio JSON and automatically synchronize new images and predictions into annotation workflows.
- Deployed a secure Label Studio platform in **AWS App Runner** with data ingested in **AWS S3**, with controlled access via **VPN**.
- Led cross-functional sessions with biologists to define annotation standards, improving label consistency and data quality.
- Engineered preprocessing pipeline to tile high-resolution images into **640×640 patches with overlap** after the annotated data were ready, preserving small-object features and reducing edge information loss.
- Applied **data augmentation** (geometric and brightness transformation) to increase dataset diversity and model robustness.
- Fine-tuned the existing **YOLOv12m** baseline model using newly annotated and augmented data. Conducted comparative evaluation across multiple architectures (**YOLO** variants, **Faster R-CNN**), identifying **YOLO26m** as marginally higher in accuracy.
- Selected the fine-tuned **YOLOv12m** model for deployment based on a balanced evaluation of **accuracy, latency, and system compatibility**, improving performance to **90% mAP@0.5.**

Technologies: Python, PyTorch, YOLO, Faster R-CNN, AWS (S3, Lambda, App Runner), Label Studio, Data Annotation, Computer Vision, Object Detection, Image Processing

Early Battery Degradation Prediction & Feasibility Analysis ***October 2025 – December 2025***

Evaluated the feasibility of early-life battery degradation prediction using long-term **time-series data**, identifying key **data limitations** and **domain gaps** affecting model generalization from lab-scale datasets to real-world battery systems.

Key Contributions:

- Analyzed **longitudinal battery lifecycle data (~1500 cycles over 10 months)** including voltage, current, capacity, and resistance signals across charge–discharge cycles.
- Built **diagnostic tools** and **data visualization** pipelines to validate cycle consistency and inspect degradation trends over time.
- Performed **anomaly detection** and **data cleaning** to remove irregular and corrupted cycles, improving dataset reliability.
- Engineered **statistical and temporal features** to capture degradation patterns across lifecycle stages.
- Determined that single-pack data lacked sufficient **population variance** for robust **supervised learning** and **generalization**.
- Trained an **XGBoost** model on the **MATR benchmark dataset** to predict end-of-life (80% capacity) from early-100-cycle features as a controlled baseline.
- Evaluated model transferability and identified **domain mismatch** between lab-scale datasets and real-world battery pack behavior. Informed project reprioritization by demonstrating that data constraints were the primary bottleneck instead of model selection.
- Explored scalable ML frameworks (BEEP, BatteryML) and modeling strategies for battery analytics prior to project reprioritization.

Technologies: Python, Pandas, NumPy, XGBoost, Time-Series Analysis, Feature Engineering, Statistical Modeling, Data Cleaning, Visualization

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Data Scientist / Data Engineer
***SynergisticIT**, Fremont, CA | June 2023 – September 2025*

Cloud-Based Clinical Workload Monitoring System ***January 2025 – August 2025***

Designed and implemented a **cloud-based data pipeline and analytics platform** for a hospital to consolidate clinical, HR, and scheduling

data for daily reporting on provider workload, documentation timing, and staffing efficiency. The system integrated **AWS-based ETL pipelines**, a **Snowflake** data warehouse, and **Power BI** dashboards to deliver near real-time insights for HR and clinical operations, enabling proactive workload balancing and reducing reporting latency by over 80%.
presenting key metrics in an interactive and user-friendly format.

Key Contributions:

- Designed and maintained a multi-stage **ETL pipeline** leveraging **AWS DMS** to capture daily CDC updates from data from multiple systems in **Amazon RDS** into a partitioned **raw S3** zone, then using **AWS Glue (PySpark)** to transform and standardize datasets into a curated **processed S3** zone for efficient loading into **Snowflake**.
- Implemented **data standardization**, **null handling**, **duplicate resolution**, and **timestamp alignment** to ensure cross-system consistency.
- Performed **SQL-based data quality check** and **remodeled** the data into **star schema** with **SCD Type 2** dimensions in Snowflake to support business-critical reporting.
- Applied **column-level masking** in Snowflake and aligned all **data access controls** with HIPAA compliance.
- Automated pipeline orchestration using **AWS Managed Workflows for Apache Airflow (MWAA)**. Monitored workflows using **AWS CloudWatch** and configured SNS alerts for pipeline failures, or data anomalies.
- Collaborated with BI analysts to define clinical workload metrics, built curated **Snowflake views**, and implemented **row-level security** for department-specific access.
- Developed **Power BI dashboards** for HR and operations teams, visualizing appointment volumes, documentation delays, off-shift activity, and workload distribution by clinic and provider role.
- Enabled HR to proactively identify staff overloads, reduce burnout risk, and improve resource allocation; cut reporting time from 12 hours to under 3 hours, supporting data-driven staffing decisions across multiple clinics.
- Trained users on pipeline and dashboard functionality and interpretation, facilitating adoption for HR team

Technologies: AWS (Glue, S3, RDS, DMS, CloudWatch, SNS, IAM, KMS, CloudFormation, MWAA/Airflow), Snowflake, SQL, PySpark, Power BI, Python

AI-Powered Social Media Sentiment Analysis and Reporting ***July 2024 – December 2024***

Partnered with a data scientist to develop an **Azure-based NLP** system for sentiment analysis on 16,000 post-sale reviews for an online clothing store. Built an **Azure Data Factory** ingestion pipeline and implemented an **Azure Databricks (PySpark)** workflow to clean, process, and classify review text using a **BERT-based model**. Visualized sentiment trends and generated automated weekly reports to support product feedback analysis and marketing decisions.

Key Contributions:

- Configured an **Azure Data Factory** pipeline to ingest data from the server API into Azure **Data Lake Storage**, scheduled for weekly trigger updates.
- Designed an **Azure Databricks** notebook using **PySpark** to perform data cleaning, deduplication and null handling.
- Implemented **NLP** models for text cleaning, lemmatization, stop-word removal. Performed sentiment intensity analysis with a **BERT** based model from **Hugging Face** Transformer. Trained a **Random Forest classifier** to predict sentiment labels.
- Evaluated model performance and optimized preprocessing parameters to improve classification accuracy.
- Visualized results by time and item, producing comprehensive sentiment trend reports. Concluded an overall satisfactory rate of 96% and trend upward over time.
- Automated the entire workflow from ingestion to reporting using **ADF triggers** and Databricks job scheduling.

Technologies: Python, PySpark, Azure Data Factory, Azure Databricks, Azure Data Lake Storage, Hugging Face Transformers, NLP, BERT, Random Forest

Predictive Sales Analytics & Forecasting Platform ***January 2024 – June 2024***

Developed a **machine learning forecasting model** to predict total sales for each product and store for the upcoming months using daily historical sales data. This involved data preprocessing, feature engineering, and applying ensamble models including **Ridge**, **XGBoost**, and **LightGBM.** The project optimized model accuracy and provided valuable forecasts for inventory management and strategic planning.

Key Contributions:

- Imported and merged multiple datasets into **pandas** DataFrames, removed duplicates and imputed missing value.
- Engineered numerical, categorical, and text-based features, including **TF-IDF** embeddings and **matrix factorization** of product and store names, lag features, and trend-based **time series indicators**.

- Conducted **Exploratory Data Analysis (EDA)**, including visualization of target distribution and time trends. Used multivariate heatmaps to analyze numerical and categorical pairings.
- Applied **mean encoding** for categorical variables and constructed ML pipelines with emsemble model of **Ridge**, **XGBoost**, and **LightGBM** regressors.
- Performed **feature selection** using **Recursive Feature Elimination with Cross-Validation** and optimized hyperparameters using Bayesian optimization to minimize **Root Mean Square Error.**
- Evaluated models with cross-validation and deployed the best-performing pipeline for ongoing forecasting.
- Predicted future outcomes and compiled comprehensive reports. Improved forecasting accuracy by 20% over baseline, reducing inventory mismatches and supporting proactive stocking decisions across multiple stores.

Technologies: Python, Scikit-Learn, Machine Learning Pipeline, NLP, TF-IDF, mean encoding, matrix factorization, Ridge Regressor, LightGBM, XGBoost, feature selection, hyperparameter optimization.

Enterprise Sales Intelligence Dashboard (Sports Corporation) ***September 2023 – December 2023***

Developed an advanced **Power BI** dashboard for an international sports corporation to analyze sales performance, discount trends, and regional success. Integrated **SQL-based** data extraction, **Power Query** transformations, and **DAX** calculations to deliver real-time, interactive insights. The dashboard was designed to facilitate data-driven decision-making by presenting key metrics in an interactive and user-friendly format.

Key Contributions:

- Queried and filtered on-premise data using **SQL**, leveraging a **star schema** data model with the Sales table at the center for optimized reporting performance.
- Cleaned and transformed datasets in **Power Query**, ensuring data accuracy and consistency.
- Developed advanced **DAX** measures for fiscal year insights, discount analysis, and update time display.
- Designed a one-page interactive **Power BI** dashboard with key metrics, including total sales, customer counts, product sales, and discount breakdown. Enabled **scheduled refresh** for dailydata updates.
- Published the dashboard to the Power BI service and implemented user access controls.
- Collaborated with stakeholders to understand business requirements and tailor dashboards to meet their needs.
- Conducted user training sessions to ensure effective use and interpretation of the dashboard.

Technologies: Microsoft SQL Server, Power BI, DAX, Power Query, Data Modeling, Business Intelligence, Data Analytics

Retail Chain Customer & Transaction Analytics ***June 2023 – August 2023***

Built a multi-page **Power BI** dashboard to analyze product sales, customer behavior, seasonal trends, and promotion effectiveness for a retail chain. Leveraged Power Query transformations, advanced DAX measures, and interactive navigation features to deliver actionable insights, enabling marketing and sales teams to optimize promotional strategies and inventory planning.

Key Contributions:

- Utilized **Power Query** to clean and transform data, including splitting and unpivoting the Product column.
- Developed advanced **DAX formulas** to create calculated columns and measures for comparative analysis.
- Created dedicated **Product Analysis** and **Customer Analysis** pages with interactive visuals (line charts, treemaps, ribbon charts) and combined them into a **Retail Analysis** page using **bookmarks** for seamless navigation.
- Ensured alignment and consistency across all dashboard pages for a cohesive user experience.
- Presented key findings to stakeholders through **PowerPoint** summaries, driving data-informed decision-making.
- Conducted stakeholder meetings to gather requirements and incorporate feedback into the dashboard design.
- Provided training and support to end-users for effective utilization and interpretation of the dashboard.

Technologies: Power BI, DAX, Power Query, Microsoft Excel, Data Visualization, Business Intelligence, Data Analytics

EDUCATION

Master of Science in Statistics, University of Chicago ***September 2022 – June 2024***

GPA: 3.73/4.0

Bachelor of Science in Mathematics, Chinese University of Hong Kong ***September 2018 – July 2022***